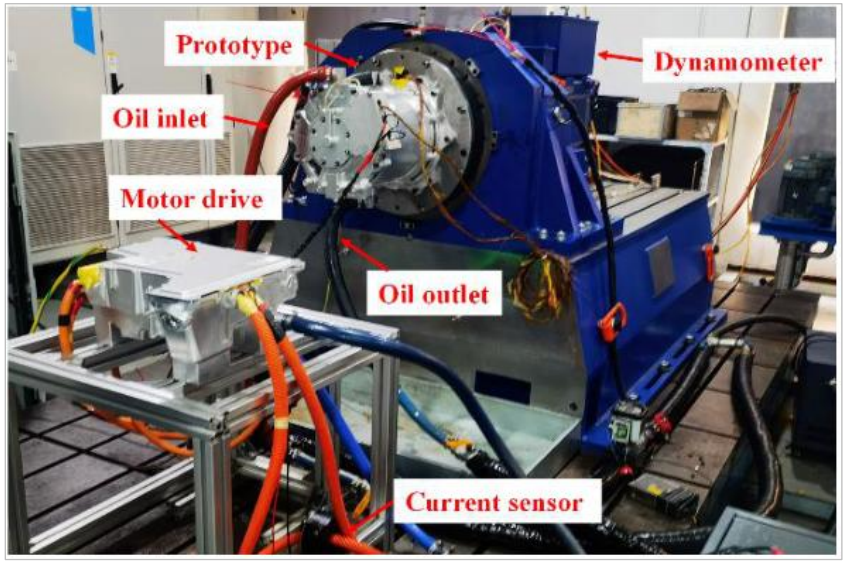
Wentao Zhang

- **Doctoral degree / institution and country / actual or expected degree date**
  PhD in Electrical Engineering, Southeast University, Nanjing, China; completed September 2025 (supervisor Prof. Zhongze Wu)
- **Other degrees (MS, BS, etc.) or relevant experience**
  M.Eng. (2021) and B.Eng. (2018) in Electrical Engineering, Southeast University, Nanjing, China
  Research Assistant, Power Electronics and Machines Centre (PEMC), University of Nottingham, UK, Sep 2025 to present (EPSRC AIMD programme, magnet-free synchronous machines)
  PhD-stage technical lead for two high-speed traction-machine prototypes in an industrial programme with Inovance
- **PhD Dissertation Title: "Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles"**
- **Awards/Honors**
  - 2025 1st Prize, IEEE Transactions on Energy Conversion Electric Vehicle Design Competition
  - 2025 2nd Prize Paper Award, IEEE Transactions on Transportation Electrification (for [J5])
  - 2025 MDPI *Machines* Best PhD Thesis Award
  - 2023 1st Prize, IEEE IAS CMD Thesis Contest; 2023 CIEEC Best Paper Award
- **Selected Publications or Research Products (Abstracts/patents, etc.)**
  26 journal papers (20 IEEE Transactions, 11 first-author), 19 conference papers, 9 patents. Selected first-author work:
  - [J5] Design of *n*-layer flat wire winding in PM machines for electric vehicles, *IEEE Trans. Transport. Electrific.*, 2025
  - [J10] High-fidelity lumped-parameter thermal model for oil-cooled hairpin winding PM machines, *IEEE Trans. Transport. Electrific.*, 2026
  - [J11] Notching and mechanical stress relieving grooves for high-speed PM machines, *IEEE Trans. Ind. Appl.*, early access, 2026
  - [C1] Deep-learning magnetic field surrogate modelling for IPM machines, ICEM 2026 (accepted)
- **Potential Notre Dame Mentor(s) and Department**
  Prof. Nishanth Gadiyar (proposed primary mentor), Department of Electrical Engineering, AMPERE Lab
- **Desired start date range at Notre Dame**
  August to October 2027

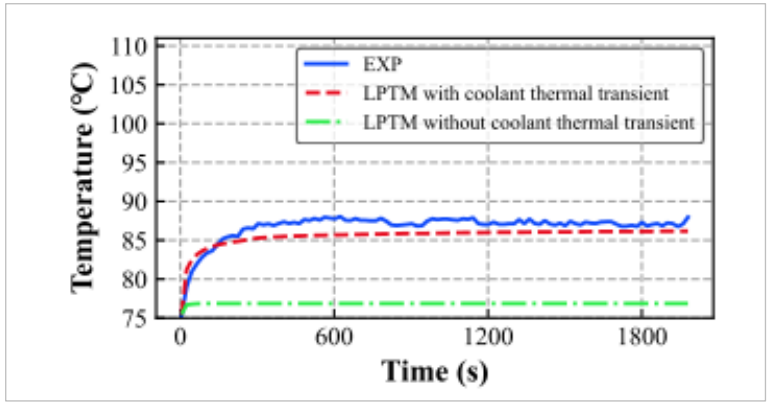
Wentao Zhang

Previous research I: multiphysics design and validated hardware

- **Previous research (include figures and citations as applicable)**
  Two high-speed oil-cooled PM traction machines taken from specification to dynamometer test in an industrial programme with Inovance (all values measured):
  – 20,000 rpm, 270 kW, 380 Nm, **7.87 kW/kg**, 97.4% peak efficiency
  – 18,000 rpm, 113 kW, 138 Nm, 7.78 kW/kg, 95.3% peak efficiency
  Models behind the hardware, each checked against test:
  – Thermal: oil-cooled thermal network **validated against thermocouples across the full operating map** [J10]; transient thermal model [J12]; thermal-network patent [P4]
  – Mechanical: rotor stress relieving grooves, confirmed by a **22,500 rpm overspeed test** [J11]
  – Electromagnetic and manufacturing: *n*-layer flat-wire (hairpin) winding methodology [J5] (2025 IEEE TTE Prize Paper Award, 2nd Prize); PWM excitation effects on loss and temperature [J22]
- **Research skills / experience**
  – Electromagnetic and structural FE: ANSYS Maxwell 2D/3D, JMAG, FEMM 4.2, ANSYS Mechanical
  – Thermal networks and Motor-CAD; SolidWorks and AutoCAD drawings with tolerances
  – Dynamometer, thermal and overspeed testing; C/C++ on TI C2000 and STM32
- **Highlight your role in project for multidisciplinary work (if applicable)**
  Technical lead from specification through design, drawing and tolerance release and supplier manufacture to AVL dynamometer test, making electromagnetic, thermal, mechanical and manufacturing decisions in one design loop.



[J11] High-speed flat-wire PM prototype on the dynamometer test rig (oil-cooled)



[J10] Measured (EXP) and thermal-network predicted temperature, oil-cooling Test II (crown section I)

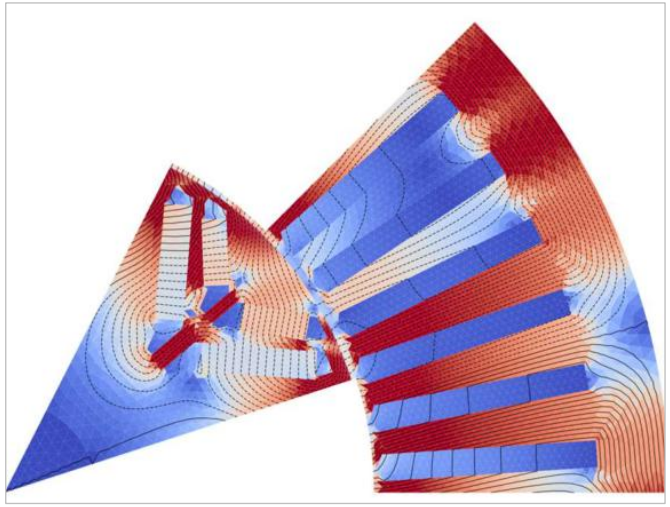
Wentao Zhang

Previous research II: physics-informed AI on an in-house FE solver

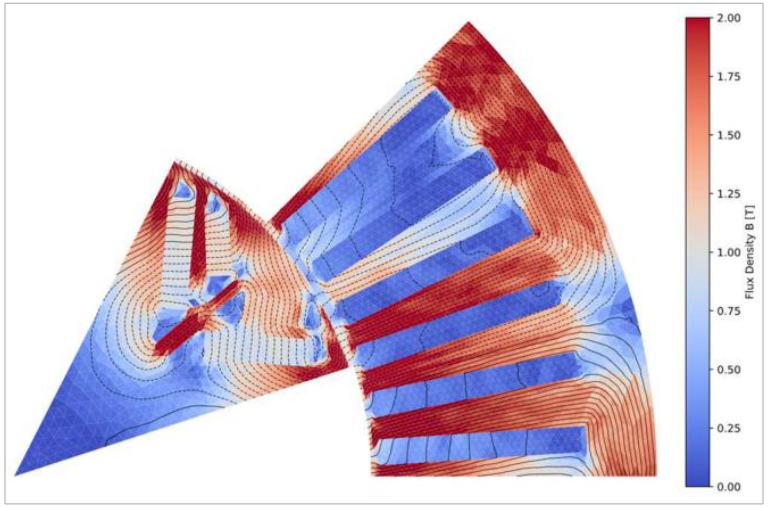
- **Previous research (include figures and citations as applicable)**
  - **In-house Python FE package** (geometry, mesh, electromagnetic, structural-modal and stress solvers), developed at PEMC and used as the reference for learned models
  - [C1] Physics-informed full-field magnetic surrogate of an IPM machine trained on this solver: R^2 = 0.9996 (torque) and 0.9730 (iron loss) on unseen designs; **more than 94% faster** than FEA; 15 design inputs
  - [J6] Hybrid AI-plus-physics optimisation cut a 67-hour FE optimisation to 2.3 hours; [P3] dual-model database patent
  - Current project: LLM-agent-assisted design and training of physics-informed surrogates (six months), supported by in-kind API credits from the OpenAI Researcher Access Program
- **Research skills / experience**
  - Python, PyTorch, MATLAB/Simulink; FE solver development
  - JMAG, ANSYS Maxwell, Motor-CAD for cross-checking learned models
  - Dynamometer testing, which supplies measured data to hold models to
- **Highlight your role in project for multidisciplinary work (if applicable)**

  First author of [C1] and [J6]; I lead the current LLM-agent project at PEMC. The work joins electrical-machine design and machine learning: I write the FE solver, train the models and check them against FE and, where available, measurement.

[C1] Flux-density field on a design not seen in training



(a) FE analysis (in-house solver)



(b) Deep-learning surrogate with Fourier layer

Wentao Zhang Proposed research

- **Proposed research (include names of potential Notre Dame mentors)**

  Physics-consistent learned models for multiphysics design of electric machines beyond rare-earth magnets

  FE evaluation of a machine design takes minutes to hours. Learned surrogates are fast, but a small field error can hide a large energy or loss error, and most are tied to one topology.

  – **Aim 1 (months 1 to 8)** Physics-consistent field operator: energy-norm and loss-consistent training; calibrated uncertainty so the FE solver is called only where the model is unsure. *Milestone:* energy-norm and iron-loss errors within a stated band on held-out designs across at least two topology families.

  – **Aim 2 (months 6 to 16)** Transfer to new topologies and materials by fine-tuning a frozen checkpoint: rare-earth-lean and magnet-free machines, AMPERE's additively manufactured SynRel and axial-flux designs. *Milestone:* transfer with a stated fraction of the FE data a from-scratch model needs.

  – **Aim 3 (months 12 to 24)** Coupled loss-thermal prediction: loss maps into the validated thermal network [J10], compared with dynamometer data. *Milestone:* validated predictions on one AMPERE machine and one Inovance-derived dataset.

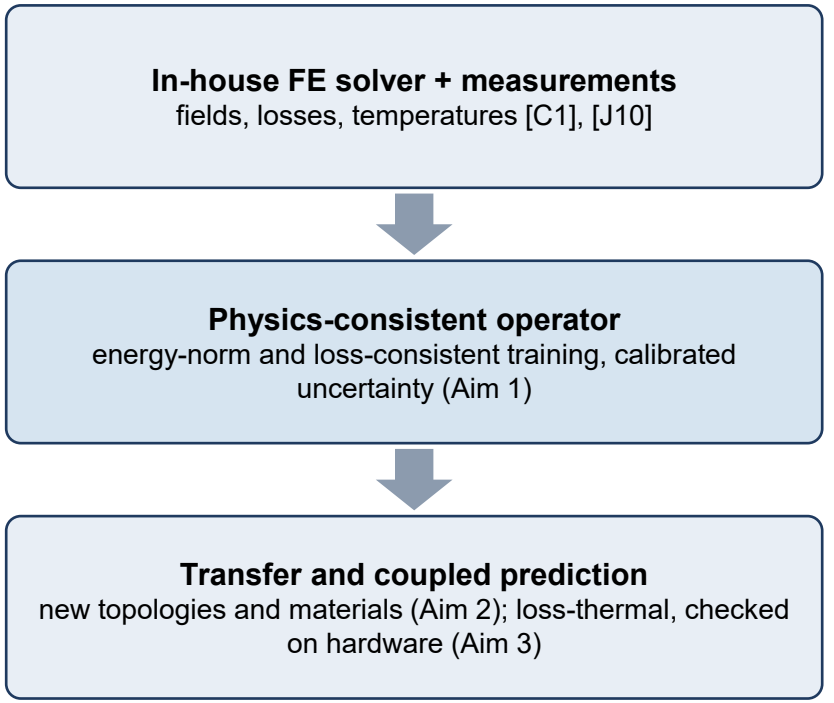
  Proposed primary mentor: Prof. Nishanth Gadiyar (Electrical Engineering, AMPERE Lab). The work connects to the lab's AI-Accelerated Design and Integrated Electro-Mechanical Systems thrusts, and its AM SynRel, axial-flux and rare-earth-lean hardware supplies the transfer test cases.

- **Resources needed for project (if applicable)**

  GPU compute from the $15,000 allowance; AMPERE Lab machines and test facilities; JMAG alongside the in-house solver.

- **Include figures and citations as appropriate**

  Schematic and preliminary results at right; citations [C1], [J10].



Preliminary, single-seed results (unpublished)

Coordinate-network / neural-operator model of the vector potential, conditioned on design and rotor position:

- FE energy-norm term: energy-norm error 29.8% → 5.4%
- Rotor-frame coherence term: rotor iron-loss error 50% → 13%
- Transfer to a new magnet grade (remanence 1.165 → 1.42 T) by fine-tuning a frozen checkpoint

Wentao Zhang Teaching, training goals and career plan

- **Teaching/mentoring/professional service experience**
  - Teaching assistant, Southeast University postgraduate course *Integrated Practice of Electrical Machine Systems* (2021 to 2025): tutorial preparation and delivery, Q&A and consultation hours, structured written feedback on final-project proposals, mark recommendations to the coordinator
  - Developed tutorial and assessment materials and two research-derived resources: a coupled electromagnetic-thermal modelling exercise and an embedded motor-drive practical
  - Day-to-day technical guidance to junior PhD and M.Eng. researchers (FEA, thermal networks, PWM-loss modelling), reflected in jointly authored papers; informal mentoring continues at PEMC
  - Referee for *IEEE Trans. Industry Applications*, *IEEE Trans. Energy Conversion* and *IEEE Trans. Transportation Electrification*
  - **Co-Organiser**, ICEM 2026 Special Session “Next-Generation Electrified Transport: Advances in Electric Drive Systems”, Funchal, Madeira, Portugal (with Dr Lingyun Shao, NUAA)
- **Goals for postdoctoral training period**
  - Build an **independent research programme** on physics-consistent learned models with the AMPERE Lab
  - Formally co-mentor graduate and undergraduate students (my mentoring so far has been informal)
  - Prepare grant proposals to NSF, DOE and ARPA-E (I have not yet led a funded project)
  - Take part in the College of Engineering Postdoctoral Training program and its cohort
  - Build a US research network through Prof. Gadiyar’s WEMPEC, GE Research and Oak Ridge National Laboratory links
- **Career plans/goals following fellowship**

  A **tenure-track faculty post** in electrical machines and physics-informed AI, leading a group that designs, builds and tests machines and holds its learned models to measurement.